
Correlated Failure as an Indirect Signature of Shared Model Provenance: A Population-Scale Complement to Instance-Level Attribution

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Contributive and corroborative attribution methods answer “which training data
2 explains this output,” one output at a time, and they need access to training data or
3 model internals that is routinely unavailable for open leaderboard submissions. We
4 describe a training-data-free, population-scale signal that is available even when
5 neither is: after conditioning on the exact, fit-free null implied by Rasch sufficiency,
6 1,228–1,362 open models per benchmark share residual failure correlation 2.8–
7 11.8 $\times$ their null, concentrated in a handful of latent dimensions rather than one
8 shared factor, and this signal survives removing exact near-duplicate submissions.
9 We could not decompose this signal against declared model lineage — a pre-
10 registered check found only 23.3% of models had a resolvable `base_model` field,
11 so we dropped the analysis rather than estimate it on a biased subset. We think
12 that failure is itself informative: it is a concrete instance of exactly the provenance-
13 opacity problem this workshop’s community is positioned to attack, and we pose
14 the residual correlation matrix as an idea-track candidate for what a black-box,
15 aggregate provenance signal could look like when nothing about training data is
16 disclosed.

17 1 The gap we are pointing at

18 Contributive attribution estimates the causal effect of specific training examples on a specific genera-
19 tion; corroborative attribution finds sources that textually support a specific output. Both operate at
20 the level of one model, one output. Neither is designed for — and as far as we know, neither has been
21 tried on — the regime of an open leaderboard with thousands of independently submitted models,
22 most of unknown or undisclosed lineage, where the question is not “which example explains this
23 generation” but “which models share unstated ancestry with which other models, and can that be seen
24 at all without the training data.”

25 We ran into this question directly, from the measurement side rather than the attribution side, while
26 trying to characterize correlated failure between open language models on standard multiple-choice
27 benchmarks. We report what we found because we think the gap between what we needed and what
28 currently exists is itself a useful data point for this workshop, even though we do not close it.

29 2 A population-scale, training-data-free residual signal

30 Let $F_{im} = 1$ iff model i fails item m . The row and column margins of F are the jointly sufficient
31 statistics of the Rasch model, so conditioning on both — sampled here with curveball (Strona et

al., 2014), verified against enumeration on small fibres — gives an exact, fit-free null under which mean-level co-failure is provably degenerate: it is a deterministic function of item difficulty alone, independent of any shared behavior between models (Schluter, 1984; Gotelli, 2000; Cronbach, 1951). At that null, on the archived Open LLM Leaderboard (five benchmarks, 1,228–1,362 models each, harvested from result files at zero inference cost), the residual correlation structure that remains after removing the difficulty degeneracy is $2.8\text{--}11.8\times$ its null value, and spectrally concentrated: on ARC-Challenge, six eigenvalues exceed the null’s spectral edge and the leading eigenvalue explains only 54% of squared residual mass, the signature of several latent dimensions rather than one shared factor. This is not a duplicate-submission artifact: clustering models at agreement 1.0000 and keeping one representative per cluster removes 900 of 1,362 ARC models and moves the calibrated participation-ratio ratio from 0.052 only to 0.058. Whatever produces this structure survives the crudest form of provenance filtering available to us.

That last clause is doing real work: it is the crudest form available, because we do not have access to a better one. We do not observe pretraining corpora, fine-tuning data, or distillation relationships for the large majority of these models. A leaderboard submission carries an optional, self-reported `base_model` field, and in our sample it was resolvable for only 23.3% of models — below a threshold we set before looking at the data specifically to decide whether a lineage decomposition was trustworthy. It fell short, so we dropped that analysis rather than report it on a biased subset. We are not claiming the residual correlation *is* shared provenance; the honest statement is narrower and, we think, more useful to this audience: standard attribution machinery cannot currently tell us either way, because it needs exactly the access this population does not provide.

3 Why this might be attribution’s problem, not measurement’s

The residual structure has three properties that make it look like a provenance signal rather than a pure behavioral coincidence. First, it is not explained by discrimination heterogeneity alone: a two-parameter item-response simulation with realistic discrimination variance produces residual correlation of at most 0.076 (rms), against an observed 0.248 on ARC. Second, it is not explained by literal duplicate submissions, as above. Third, its dimensionality is stable and benchmark-dependent (4–6 latent dimensions on four benchmarks, 12 on HellaSwag), which is the kind of structure one would expect from a handful of dominant pretraining lineages or shared distillation teachers producing correlated errors along a few behavioral axes, rather than from either a single dominant lineage or from fully independent training runs.

None of this is proof of shared provenance — discrimination heterogeneity, unmodeled ability dimensions, and genuinely convergent training choices are alternative explanations we cannot rule out with this design, and our own limitations section says so plainly. What we can say is that the exact conditional null gives a population-scale test that is exactly invariant to item difficulty by construction (Lemma 1), costs no training-data access, and is sensitive to structure at a scale (thousands of models) where per-example causal attribution is not currently run at all.

Lemma 1 (Mean co-failure depends only on item margins) $O = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$, where c_m is the number of models failing item m ; hence any randomization preserving item margins leaves O exactly invariant.

4 An idea-track question

We pose this as a question rather than a method: can the residual eigenvectors of a margin-conditioned co-failure matrix be matched — even loosely, even just as a screening tool — against known facts about model families, base-model announcements, or public distillation claims, to turn an aggregate statistical signal into something closer to an attribution claim? If so, this would be a training-data-free, black-box complement to contributive and corroborative methods, useful specifically in the regime where those methods’ preconditions (access to training data, identified individual outputs to explain) do not hold, such as regulatory or journalistic audits of a leaderboard’s actual diversity of origin. If not — if the residual structure is better explained by convergent unimodal training choices than by shared lineage, which our own spectral diagnostics do not rule out — that would itself be a useful negative result about how much population-scale behavioral correlation the field should be willing

83 to read as a provenance signal at all. We do not know which is true, and we think that is the right
84 question for this venue rather than for a measurement paper on its own.

85 **5 Limitations**

86 We did not attempt any form of causal or corroborative attribution ourselves; this paper reports a
87 measurement and a gap, not an attribution method. Discrimination heterogeneity and unmodeled
88 ability dimensions are alternative explanations for the residual structure that our design cannot rule
89 out. Claims are restricted to the evaluated open-model ecosystem on multiple-choice benchmarks.
90 Code, data, and the pre-registration are archived and linked in the full paper.

91 **References**

- 92 L. J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16:297–334, 1951.
- 93 N. J. Gotelli. Null model analysis of species co-occurrence patterns. *Ecology*, 81(9):2606–2621, 2000.
- 94 D. Schluter. A variance test for detecting species associations, with some example applications. *Ecology*,
95 65:998–1005, 1984.
- 96 G. Strona, D. Nappo, F. Boccacci, S. Fattorini, and J. San-Miguel-Ayanz. A fast and unbiased procedure to
97 randomize ecological binary matrices with fixed row and column totals. *Nature Communications*, 5:4114,
98 2014.